

VAIBHAV KUMAR RAI

Data Analyst | Business Analyst | HR Analyst | Immediately Available

[Portfolio](#) | Vaibhavrai769@gmail.com | (+91) 9918795919 | github.com/Vaibhavrai-007 | linkedin.com/in/vaibhav-kumar-rai

PROFESSIONAL SUMMARY

Data & Business Analyst with an MCA from Tezpur University (Central University, CGPA: 7.99) and hands-on experience as a Data Science Intern at NIELIT, Govt. of India, working across SQL, Python, Excel, Power BI, and Tableau to analyze data, build dashboards, and turn insights into decision-ready recommendations. Built and deployed a machine learning regression model on 19,000+ real-estate records that cut agent-led overpricing by 15%, an executive e-commerce analysis across \$1.51M in GMV, and an HR attrition analytics dashboard on 1,400+ employee records. Experienced in requirements gathering, business case development, and stakeholder communication, with a strong track record of data storytelling for technical and leadership audiences. Open to full-time Business Analyst, Data Analyst, HR Analyst, and Data Science roles.

TECHNICAL SKILLS

- Data Analytics & BI:** Data Analysis, Advanced SQL, Advanced Excel (Pivot Tables, VLOOKUP/XLOOKUP, Macros, Power Query, Power Pivot), DAX, Power BI, Tableau, Data Visualization, Data Modeling, Data Warehousing, ETL, Dashboard Design, Data Reporting, KPI Tracking, EDA, Data Storytelling
- Programming & Machine Learning:** Python (Pandas, NumPy, Scikit-learn), Statistics (Inferential Statistics, Hypothesis Testing, A/B Testing), XGBoost, Random Forest, Regression, Classification, NLP, LLMs, TensorFlow, PyTorch
- HR Analytics:** HR Metrics & KPIs, Employee Attrition Modeling, Turnover & Retention Analysis, Workforce Analytics & Headcount Planning, Compensation Parity Analysis, CIPD & SHRM Frameworks
- Business & Soft Skills:** Business Analysis, Requirements Gathering, BRD/FRD Documentation, Business Case Development, Stakeholder Communication, Agile/Scrum, Cross-Functional Collaboration, Data-Driven Decision Making

EXPERIENCE

Data Science Intern — NIELIT Janakpuri, Delhi Centre (Govt. of India)

Feb 2026 – Aug 2026

NIELIT Smart Bridge — AI Translation & Exam Platform | [GitHub](#)

- Built a neural translation engine using NLLB-200, fine-tuned with LoRA/PEFT adapters for 8 languages and merged into a standalone model, with INT8 quantization, LRU + persistent JSON translation caching, and Indic/CJK transliteration, exposed via a Flask API with a React/Vite frontend.
- Developed a CBT exam platform (React, Vite, Supabase/PostgreSQL) with self-authored MCQ question banks, a question palette, anti-cheat hooks, timed assessments, and a Pyodide WebAssembly sandbox for practical Python exams.
- Built offline resilience using LocalStorage to auto-save in-progress quizzes during network loss and resume on reconnect, plus IndexedDB for large offline question banks; added an XP/streak-based leaderboard ranking students within a course and across all courses.
- Created an admin panel for student roster management, batch CRUD, bulk question upload, and audit logging; taught Advanced Excel, Power BI & SQL as Faculty for data analytics batches, delivering weekly stakeholder insights.

PROJECTS | [View all projects on Portfolio](#)

Global E-Commerce Revenue Intelligence — [GitHub](#)

Python, Pandas, Power BI, EDA, Pareto Analysis, Data Storytelling

- Conducted executive portfolio analysis across \$1.51M in gross merchandise value (GMV), 1,200 orders, 10 international markets, and 8 product categories.
- Implemented Pareto 80/20 concentration modeling showing robust diversification, with 7 of 10 markets driving core revenue; identified Home Decor as top category (\$232.3K gross volume, \$1,442 AOV) and peak weekly velocity on Wednesdays (\$277K).

HR Attrition & Workforce Retention Analytics Dashboard — [GitHub](#)

Excel, Power BI, DAX, Data Modeling, HR Analytics

- Conducted end-to-end people analytics on 1,400+ IBM HR employee records to identify key drivers of workplace turnover (burnout at 30.5% attrition under overtime; Sales Reps at 39.8%).
- Designed an interactive multi-page Power BI dashboard with dynamic DAX measures (Attrition Rate, Retention %, Headcount Trends) and custom cohort binning (Tenure, Age); delivered retention recommendations targeting overtime caps, compensation reviews, and early-career engagement, with an estimated 12–15% turnover reduction.

Gurgaon House Price Prediction — [GitHub](#)

Python, Pandas, Scikit-learn, XGBoost, EDA, Data Storytelling | [Live Demo](#)

- Analyzed 19,000+ Gurgaon property listings, engineered features, and tuned XGBoost/Random Forest models — cutting agent-led overpricing by 15% and saving buyers up to ₹10 Lakhs per transaction across 500+ valuations.

Weather Intelligence & Climate Analytics Platform (WICAP) — Personal Project

Ongoing

Python, SQL, ETL, Data Warehousing, Data Quality Framework, Dashboards

- Architecting an end-to-end weather/climate analytics platform (ingestion, ETL, warehousing, dashboards) with a data quality framework and trend/seasonal/anomaly-detection views filterable by geography and time.

EDUCATION

MCA — Tezpur University (Central University, NAAC A+ Grade)

2024 – 2026 | CGPA: 7.99/10

BCA — SMS Varanasi (MGKVP)

2021 – 2024 | CGPA: 7.98/10

Class XII, CBSE

2021

CERTIFICATIONS, ACHIEVEMENTS & ADDITIONAL | [View all certificates on Portfolio](#)

- NPTEL Certifications:** Data Analytics with Python — IIT Roorkee (73%) | Joy of Computing using Python — IIT Madras (71%) | Understanding Incubation & Entrepreneurship — IIT Bombay (90%)
- HR & Business Certifications:** Human Resources Foundations — CIPD & LinkedIn Learning | HR Foundations: Redefining HR — LinkedIn Learning & SHRM | Understanding Organizations and the Role of HR — LinkedIn Learning
- Data Certifications:** Career Essentials in Data Analysis — Microsoft & LinkedIn Learning | Data Visualization for Data Analysts and Analytics — LinkedIn Learning | Data Science 101 — HP LIFE | The Ultimate Job Ready Data Science Course — CodeWithHarry
- Availability:** Immediate | **Languages:** English, Hindi
- Core Strengths:** Data Storytelling, Dashboard Design, KPI Tracking, Stakeholder Communication, Business Insight Generation